Deadweight Losses or Gains from In-kind Transfers: Experimental Evidence

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Abstract

Are in-kind transfers associated with deadweight losses? To answer this question, we conducted an incentivized field experiment in India, offering low-income respondents a choice between a free quantity of rice and varying amounts of cash to elicit their willingness to pay for rice. Contrary to standard economic theory, we find evidence of deadweight gain on average, though with variation across respondents. We investigate potential explanations for this result, focusing on infra-marginality, commitment considerations and intra-household bargaining dynamics. Notably, we uncover a striking contrast: women in female-headed households exhibit deadweight losses, while women in male-headed households experience deadweight gains. Upon examining alternative explanations, our results highlight that greater bargaining power of women within households increases the propensity to choose cash over rice.

Keywords: deadweight loss, in-kind transfer, cash transfer, food subsidy, field experiment, India

JEL codes: C93, D13, I38, J16, Q18

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1 Introduction

Supporting the poor is a central concern of the modern welfare state. There are essentially two ways to help those who cannot afford to meet their basic needs. One is to provide basic goods to eligible poor households for free or at a subsidized price. Examples are food or fuel subsidies, free textbooks, and social housing. The other is to support the poor household's income directly with cash payments, which enables the beneficiaries to supply themselves with the goods they want to buy from the market. This paper examines the choice between in-kind benefits and cash transfers in an incentivized experimental setting through the novel perspective of deadweight gains or losses.

Are cash transfers a better way to support poor households than in-kind benefits? Standard economic theory offers some insights. Suppose a household consumes a certain amount of a basic good (say, grains) at the market price p_m. Let's assume that the government now offers the household the option of buying up to $\bar{x}_s$ units of this basic good at a subsidized price p_s which is below the market price p_m. Given this option, the household decides to buy x units of the basic good, of which $x_s(<= \bar{x}_s)$ units are bought at the subsidized price. Now imagine that the government cancels the food subsidy and pays the household exactly the same amount in cash. The amount of equivalent cash transfer is thus $s = (p_m - p_s)\bar{x}_s$. For the household, there are two possibilities. First, consider a household whose total consumption x exceeds $\bar{x}_s$. This household buys $\bar{x}_s$ units at the subsidized price p_s and the remaining $(x - \bar{x}_s)$ units at the market price p_m. This represents an infra-marginal household that will be indifferent between the in-kind subsidy and an equivalent cash transfer because, with an equivalent transfer, such a household continues to consume x units as under the in-kind subsidy option. In other words, an infra-marginal household's choices are not constrained by the in-kind subsidy relative to an equivalent cash transfer. Second, consider a household whose total consumption x is less than or equal to $\bar{x}_s$. This household will purchase all its consumption at the subsidized price p_s. Such a household represents a marginal household that can always do better with an equivalent cash transfer because they would then be able to spend the money according to their preferences. For marginal households, there is thus an excess burden or deadweight loss associated with the in-kind transfer.

The deadweight loss can be conceptualized as follows: a marginal household would also be willing to accept a cash transfer s^* smaller than s that makes them exactly indifferent between a cash transfer of s^* and the in-kind subsidy. The difference between s and s^* is the in-kind subsidy's deadweight loss. Since infra-marginal households experience zero deadweight loss, the two possibilities taken together imply that equivalent cash transfers should always be at least as good as, if not better for households, than in-kind transfers. Moreover, in practice, policymakers may also find cash transfers more appealing due to their lower administrative and delivery costs. For example, Hidrobo et al. (2014) estimate that delivering food transfers in Ecuador is three times more expensive than distributing cash transfers.

Despite the strong case against in-kind benefits, they remain widespread in social protection programs across the globe for many reasons (Alderman et al., 2017; Currie and Gahvari, 2008;

Gadenne et al., 2024). Sometimes there are paternalistic considerations, e.g. the provision of free textbooks aims to ensure that the children benefit from the subsidy by making it harder for parents to divert an equivalent cash transfer for other purposes. Sometimes equivalent cash transfers are politically or administratively unfeasible. Moreover, in contexts where in-kind transfers have been implemented at scale, the evidence suggests that most beneficiaries are infra-marginal. For instance, for the Public Distribution System in India – the largest food subsidy program in the world with a reach of about 900 million people – it is estimated that 93% of beneficiaries are infra-marginal (Gadenne et al., 2024). Similarly, for the Rasta Program in Indonesia, 97% of beneficiaries are estimated to be infra-marginal (Banerjee et al., 2023). Thus, it seems that deadweight losses of in-kind transfers anticipated in standard theory need not be a major concern in practice.

There can however be several grounds for expecting departures from the standard theoretical model. A range of additional factors may influence beneficiaries' choice between in-kind or cash transfers, including: differences in transaction costs or trust in the implementation of the cash and in-kind modalities; differences in the quality of market and subsidized goods; use of the in-kind transfer as a commitment device by beneficiaries to avoid the possibility of cash from being spent on other things; the insurance value of in-kind transfer against market price volatility; and the asymmetry of bargaining power within the household.

Our focus in this paper is to test the existence of deadweight loss of in-kind benefits, in light of the above considerations influencing household behavior. We quantify the magnitude of deadweight loss through an incentivized field experiment. Such measurement of deadweight loss has not received much attention in the literature. Our experiment is located in the context of the Public Distribution System in India, which has been in operation for decades. We conducted the experiment in selected low-income urban neighborhoods in the state of Maharashtra, where we offered our predominantly female respondents the choice between a cash transfer and a fixed quantity of rice provided free. Using an incentive compatible procedure, we elicited the respondents' willingness to pay for rice (the minimum amount of cash they would be willing to accept instead of the rice), which allowed us to quantify the deadweight loss associated with the food subsidy.

Our headline result shows that the willingness to pay for rice was predominantly above the market value of the rice, thus documenting an overall deadweight gain rather than a deadweight loss from in-kind benefits. While this stands in contrast to the prediction from standard economic theory, several additional factors, as noted above, may be relevant to this finding. Our experiment enabled us to investigate these factors influencing respondent choice. In particular, the design and the setting of our experiment allows us to rule out the potential role of transactions costs, quality of the product, trust considerations and insurance against price volatility. Our data suggests that the explanation for our headline result can instead be found in the role of infra-marginality, the use of rice as a commitment device and intra-household bargaining. We find that controlling for infra-marginality, there are systematic differences along the dimension of bargaining power differentials within the household. Using male/female headship as a proxy for bargaining power differential, we find that female respondents from female-headed households make choices entailing deadweight losses in contrast to choices of female respondents from male-headed households which entail deadweight gains.

We also find evidence for the use of rice as a commitment device by our respondents. However, the proportion reporting this as the primary reason for their choice is significantly higher in male- than female-headed households. We argue that the greater (reported) need for a commitment device among women from male-headed households further points to their relatively lower bargaining power.

Our study contributes to three main strands of the literature. First, it contributes to the literature on measuring deadweight loss of in-kind transfers or gifts. Using evidence from surveys, Waldfogel (1993), and Principe and Eisenhauer (2009) show that gift giving can lead to a deadweight loss, reflecting the sub-optimal nature of the gift selected by the gift-giver that does not match the recipient's preferences. On the other hand, surveys conducted by Solnick and Hemenway (1996) show evidence of substantial deadweight gain as respondents appreciated the thought that went into choosing a gift. List and Shogren (1998) compare survey-based results with incentivized auctions and report a deadweight loss using surveys and a modest deadweight gain using incentivized methods. Cunha et al. (2019) study the welfare effects of in-kind transfers, operating through a price effect, in the context of a program in Mexico that randomly assigned in-kind transfers, equivalently-valued cash, or no transfers to villages in the sample. They find that in-kind transfers increased supply of the good in the recipient community, reduced price, and had a substantial (positive) welfare impact on poor villages. In contrast to these studies, our paper takes a different approach. We conduct an incentivized field experiment where households are offered a choice between different amounts of cash and a fixed in-kind transfer; the point at which they switch from in-kind to cash allows us to construct a more direct measure of their deadweight loss.

Second, our study connects with the debates around the Public Distribution System (PDS) in India, the world's largest safety net program based on in-kind transfers of highly subsidized food (mainly wheat and rice) with an estimated coverage of about two-thirds of the country's population or nearly 900 million people (Khera and Somanchi, 2020). Historically, a major concern with the PDS has been the diversion of subsidized grains to the open market, with the estimates of such "leakage" ranging between 35 and 47 percent of the total grain released for distribution for 2011-12 (Drèze and Khera, 2015; Gulati and Saini, 2015; Himanshu and Sen, 2013; Jha and Ramaswami, 2012). This has led to calls for reforms to introduce the option of direct cash transfers in lieu of in-kind subsidy (Basu, 2011; Kotwal et al., 2011; Muralidharan et al., 2019). Survey-based evidence from beneficiaries, on the other hand, often reports a preference for non-cash transfers over cash transfers when presented with hypothetical choices between the two (Khera, 2011, 2014; Muralidharan et al., 2011; Satapathy et al., 2022) or in the context of pilot programs that rolled out direct cash transfers to replace subsidized food (Muralidharan et al., 2017).[1] However, incentivized experimental evidence on this important

[1]Survey-based evidence from Muralidharan et al. (2011) reports that the minimum value of cash for which respondents were willing to forgo their food ration was higher, on average, than the value of the food subsidy they were receiving. Hirvonen and Hoddinott (2021) use Ethiopian survey data on beneficiary preferences between cash and in-kind and find that most beneficiaries stated they prefer their payments only or partly in food.

policy issue is rare. One recent exception is Khanal et al. (2024) who conduct a partially incentivized two-step experiment giving respondents in Nepal a choice between cash and an equivalent value of rice or fertilizers, but they do not elicit willingness to pay for the in-kind transfer and hence are unable to provide evidence on deadweight losses or gains. The second step of their experiment, which considers three higher values of in-kind transfers, is not incentivized.

Third, our paper also advances the research on decision-making within households. Starting with the important contribution of Chiappori (1992), researchers have used theoretical models, observational studies, experimental games and impact evaluations to understand intra-household behavior. Baland and Ziparo (2018), Chiappori and Mazzocco (2017), Munro (2018) and Doss and Quisumbing (2020) present useful surveys of this literature. The main themes in this literature revolve around the idea that, while many resources are owned and managed jointly by household members, and several decisions are made jointly, not all parties necessarily have equal voice in these decisions. Further, households do not always reach efficient outcomes. For instance, for the United States, a higher propensity for food consumption out of food stamps than out of cash income has been explained in terms of intra-household allocations when there are multiple earning members, but only one of them contributes to food spending (Breunig and Dasgupta, 2005). Closer to the context of our study, survey evidence suggests that heterogeneity in terms of class, caste, gender and political affiliation can influence preferences for the delivery mechanism of the food support system in India (Khera, 2014; Pradhan et al., 2019). In contrast to this literature, we focus on eliciting willingness to pay for in-kind benefits using female or male headship as a "distribution factor" (Browning et al., 2013). This approach provides an insight into how bargaining power differences influence household choices and the resulting deadweight loss or gain.

2 The Experiment

This section provides details of the sample, the baseline survey, and the experiment. The experiment was conducted in low-income urban neighborhoods (hereafter "slums") of Nashik, a city in the western state of Maharashtra, India. The questionnaire modules and experiment rounds were designed and implemented using the World Bank's Survey Solution suite of Computer Assisted Personal Interview (CAPI) software system.[2]

To identify the survey slums, we first extracted the list of all the slums in Nashik from the Census of India 2011 and randomly selected 10 slums from that list. For the selection of respondents in each slum, the following procedure was used. We conducted a listing operation where approximately 100 households were selected in each slum using a random route method, where for a slum with N total number of households, the surveyors walked around the slum and listed every (100/N)-th household. Table A1 in the Appendix A provides the details of the

Similarly, Ghatak et al. (2016) using survey data from India find that a majority of beneficiaries prefer receiving in-kind benefits (bicycles) rather than cash transfers. They find the preference to be correlated with household characteristics such as income and household size.

[2]Survey Solutions is an open-source software designed and developed by the World Bank that has been used extensively to conduct household surveys around the world.

listing operation. The listing operation provided us with the sample frame for each slum. We then drew a random sample of 25 households from each slum. The resulting 250 households constituted the final sample for our experiment.

One individual was identified as the respondent from each of the 250 households. The surveyors were instructed to identify the member who usually made rice purchases for the household and ask that person to participate in the experiment. If such a person was absent, then they were instructed to identify any other adult. Unsurprising for our setting, nearly 90% of our respondents were women. We do not randomize the gender of the respondents to ensure that our experiment captures the prevailing decision-making environment in relation to food spending.

A baseline survey was conducted prior to the experiment which collected detailed information for sample households, including: the household's social group and religion; member characteristics such as age, gender, relationship to the head, marital status, educational attainment, employment status, major source of income, disability if any; details about their dwellings and asset ownership; their grain purchases both from the public distribution system and the market; weekly purchase of food items and their prices; details of bank accounts and their usage; and decision-making within the household.

The experiment consisted of three rounds in which respondents were offered a choice between a fixed quantity of rice and varying amounts of cash ranging between values both below and above the market value of the rice. At the time of the experiment the going price of rice in the local markets was Rs 32 per kilogram with no significant variation across slums. There was also very little temporal variation in rice prices over the period of the experiment, March-August 2019.

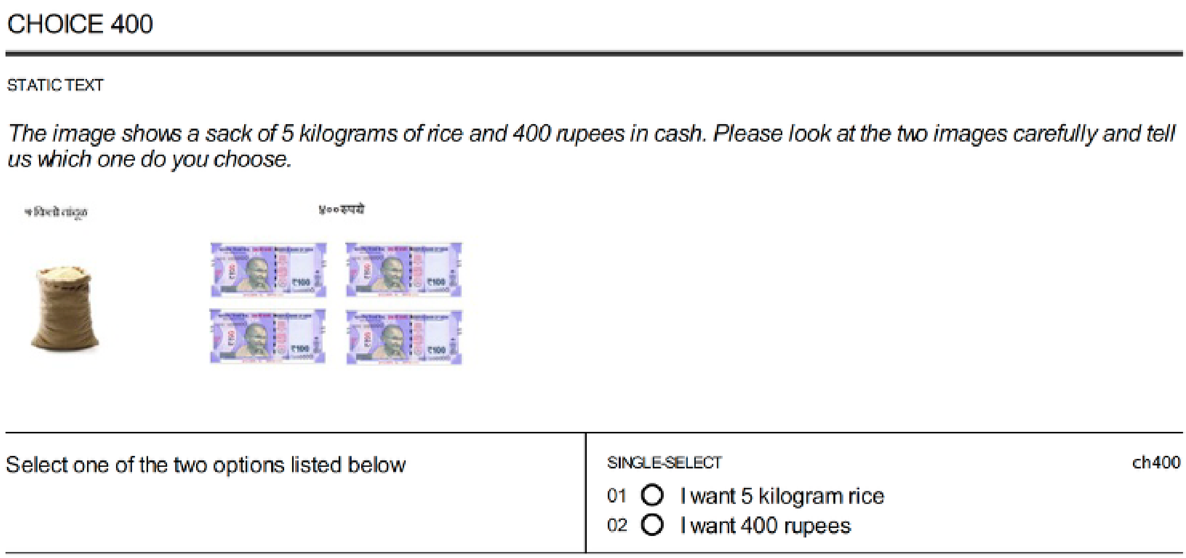
Hence, in each round of the experiment, respondents were offered choices between 5 kilograms of rice and nine alternative cash amounts. The choices ranged from the lowest cash value of Rs 50 increasing thereafter in 50 rupee increments up to Rs 400 and a final choice with the highest value of Rs 500. Given the market value of Rs 160 (for 5 kilograms of rice), the end points of this range were selected to ensure that there was an unambiguous incentive to choose rice (cash) at the lower (upper) bound. In total, therefore, the respondents were offered nine choices which correspond to our nine treatments. The specific parameters for the cash offers were based on data from a pilot run. While there was a natural lower bound of zero for cash offers, the pilot experimented with values of the upper bound below Rs 500. The pilot data showed a large number of respondents always choosing rice at these upper bounds, which was the main justification for raising the upper bound of the cash offers up to Rs. 500 (approximately three times the market value of 5 kilograms of rice) in the final design.

As the respondents were offered increasing cash amounts against 5 kilograms of rice, we expect the cash option to become increasingly attractive. A key aim of the experiment was to identify for each respondent the switch point where the cash option becomes preferable. This switch point offers us a measure of the respondent's willingness to pay (WTP) for 5 kilograms of rice, which in turn enables us to measure deadweight loss (DWL) as DWL=160-WTP. It is important to note that the WTP measure is not based on hypothetical scenarios and is instead incentivized in the experiment by randomly selecting one of the nine respondent choices for redemption. Specifically, the surveyor read the following statement explaining the experiment to the respondent.

> "*We will now be asking you to make choices between receiving 5 kilograms of rice or receiving different amounts of cash. We will be asking you to make a choice nine times. Each time, we will be asking you to tell us whether you would prefer to get a particular amount of cash or 5 kilograms of rice. Please choose carefully because these are not just hypothetical choices. Later, one of these choices will become real when you draw a number from the lottery bag. The number you draw from the bag will tell us which of the nine choices is selected, and that will determine what you will get. For example, if you picked the number 300, then we will look at your preference between 300 rupees and 5 kilograms of rice, and if you had chosen 5 kilograms of rice, you will get 5 kilograms of rice, not 300 rupees. Or instead, if you had chosen 300 rupees, you will get 300 rupees, not 5 kilograms of rice. So, your choices will matter to what you can get. Hence, make your choice thoughtfully. So, let's now begin by asking you about your choices. Please note that at the current market price of about Rs 32 per kilogram of rice, the value of 5 kilograms of rice is about 160 rupees.*"

After this statement was read out, the respondents were shown the nine cash versus rice choices one by one on the tablet. Figure 1 presents an example of the choice question shown to the respondent. Once all the choices were recorded, the respondent was asked to draw a number written on a piece of paper from a lottery bag, which contained nine pieces of paper each bearing one of the nine cash amounts. For the number drawn by the respondent, their previously recorded choice was selected for redemption. For example, if the respondent drew the number 250 and for the choice option of Rs 250 versus rice the respondent had chosen rice, then the respondent was given a voucher for 5 kilograms of rice; otherwise, they were given a voucher for Rs 250.

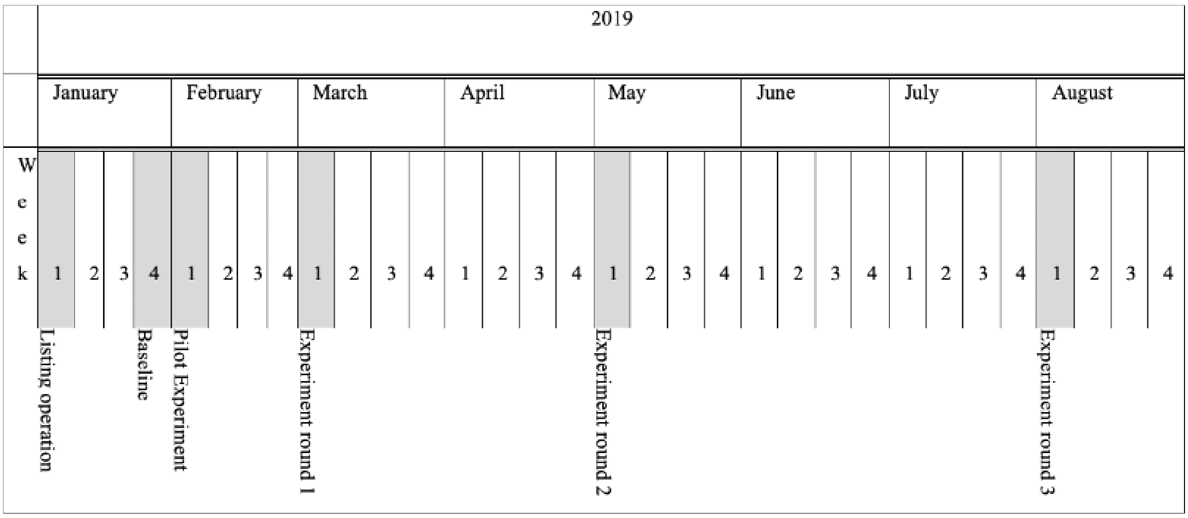
Figure 1: Snapshot of the choice question



Notably, respondents were given a voucher for either choice - both five kilograms of rice or their chosen cash amount - to be redeemed at exactly the same location, viz., at their slum's local shopkeeper. We selected one shopkeeper in each slum who was assigned the task of disbursing rice or cash to the selected respondents upon the presentation of the voucher. The shopkeepers were instructed to first match the voucher number in the respondent list that we provided to them, and then distribute cash or grain as indicated on the voucher. Figure A1 shows images of the vouchers and the surveyors interviewing the respondents using the tablet.[3] We verified that all vouchers were successfully redeemed.

The timeline of the experiment is presented in Figure 2. The listing operation was completed and the final sample was selected by the second week of January 2019. A baseline survey was completed in the last week of January 2019. The pilot was conducted in the first week of February 2019, and the experiment procedures were revised based on the experience with the pilot. Three rounds of the experiment were conducted in March, May and August 2019 during the first week of each month.

Figure 2: Timeline of the experiment



3 Headline Results

For our sample, nearly 90% of the respondents were female. This is as expected since our experiment targeted adult members responsible for food purchases who are mostly women in our setting. Table A2 presents summary statistics for the female respondent and the full samples. The average age of respondents was 37 years in both the full sample and the sub-sample of female respondents. About 29% (27%) of the households were female-headed in the female respondent (full) sample. We present our results for the sample of female respondents so as to mitigate any confounding effects related to the respondent's gender. Appendix B shows the corresponding results for the full sample which are very similar in magnitude and significance.

Table 1 presents the distribution of respondent choices against each of the cash offers pooled

[3]The full set of experimental instructions are shown in Appendix E.

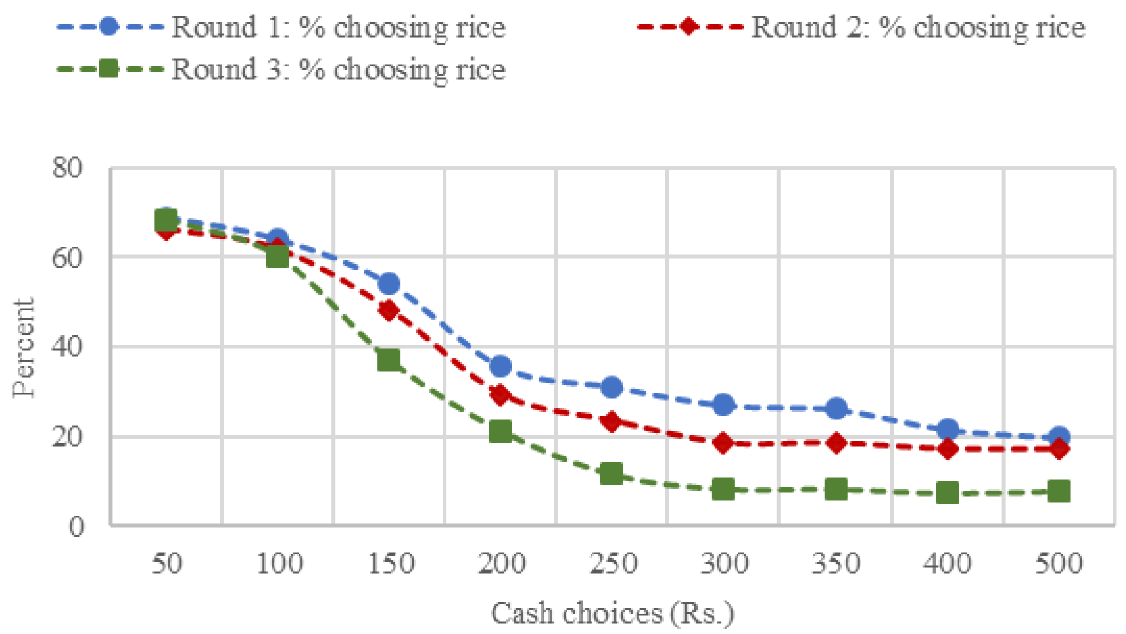
over all three rounds. As expected, at higher cash amounts, a lower proportion of households opt for rice rather than cash, ranging from 68% choosing rice when offered the minimum amount of Rs 50 to 15% when offered the maximum of Rs 500. This pattern holds for all three rounds (Figure 3).

Table 1: Percentage of respondents choosing rice against each cash offer, pooled across rounds

Cash offer (Rs.)	50	100	150	200	250	300	350	400	500
% choosing rice	67.9	62.0	46.5	28.8	22.1	18.0	17.7	15.5	15.0

Notes: The sample for this and all subsequent results reported in the main text relates to female respondents only. Appendix B reports the corresponding results for the full sample.

Figure 3: Percentage of respondents choosing rice against each cash offer across the three rounds



Based on the type of choices made by respondents, we can distinguish three types: first, the "single-switch respondents" as those who made a single switch from rice to cash as higher cash amounts were offered; second, "rice-only respondents" as those who chose rice for all nine cash offers; and third, "cash-only respondents" as those who always chose cash. There were also a small number of respondents (3.5% of the pooled sample) who switched multiple times between rice and cash.[4] Multiple switches are hard to interpret and hence we exclude these respondents from our analysis. Thus, our final pooled sample consists of a panel with 660 observations (222, 222 and 216 respondents from rounds 1, 2 and 3 respectively). The proportions of single-switch, rice-only and cash-only respondents in the final pooled sample are 54, 14 and 33 percent

[4]Multiple switches are not uncommon in similar elicitation methods. For instance, in case of risk elicitation, multiple switches are reported for 8.5% of the subjects in Dave et al. (2010), 13% in Holt and Laury (2002) and over 50% in Jacobson and Petrie (2009) and Charness and Viceisza (2016).

respectively (Table 2).

Table 2: Number and percentage of cases for each type of respondent

Respondent type	Single-switch	Rice-only	Cash-only	>1 switch	Total
Round 1					
Number	101	38	67	16	222
% of sample*	45.5	17.1	30.2	7.2	100
% of final sample	49.0	18.5	32.5	—	100
Round 2					
Number	109	36	75	2	222
% of sample*	49.1	16.2	33.8	0.9	100
% of final sample	49.6	16.4	34.1	—	100
Round 3					
Number	131	14	66	5	216
% of sample*	60.7	6.5	30.6	2.3	100
% of final sample	62.1	6.6	31.3	—	100
Combined					
Number	341	88	208	23	660
% of sample*	51.7	13.3	31.5	3.5	100
% of final sample	53.5	13.8	32.7	—	100

Notes: * Including respondents with multiple switches, while the final sample excludes them. For instance, the final combined sample comprises 637 cases (660 minus 23 multiple-switch cases).

While WTP is in principle a continuous variable that cannot be elicited in an experiment with a finite number of treatments, our experiment allows us to observe an interval containing the WTP. For example, if a single-switch respondent opted for rice at Rs 100 but switched to cash at Rs 150, then its WTP lies in the switch interval [100, 150]. On the other hand, for rice-only respondents, their WTP is bounded below by Rs 500. Similarly, the WTP for cash-only respondents is bounded above by Rs 50. We thus construct our measure of WTP for the three types of respondents as follows: we approximate the WTP for single-switch respondents as the midpoint of their switch interval; for rice-only respondents, we assume their WTP to be Rs 550; and the WTP for cash-only respondents is assumed to be Rs 25. The deadweight loss for a respondent can then be defined as

$$\text{DWL} = \begin{cases} (160 - \text{WTP} & \text{if single-switch} \\ (160 - 550) & \text{if rice-only} \\ (160 - 25) & \text{if cash-only} \end{cases}$$

where Rs 160 is the market value of 5 kilograms of rice. The deadweight loss is thus not necessarily always positive; a negative value of DWL indicates that the respondent's willingness to pay for rice exceeds its market value.

Table 3 presents the estimates of WTP and DWL by respondent type. By construction, DWL for cash-only respondents is positive and that for rice-only respondents is negative. It also turns out that cash-only respondents account for more than twice as many cases as rice-only respondents. However, since the positive DWL for the former (135) is dominated by the negative DWL for the latter (-390), the combined average DWL for these two types of nonswitching respondents is notably negative. A more striking result is that the average DWL for single-switch respondents is also negative with an estimate of Rs. -17 (statistically different from zero with a p-value of 0.0001). This implies that for our sample as a whole, the average DWL is negative, i.e., a deadweight gain (DWG) overall, with a point estimate of Rs. 19 (p-value of 0.007).

Table 3: Distribution of Willingness to Pay (WTP) and Deadweight Loss (DWL) by Respondent Type

Respondent type	Switch interval	WTP (Rs.)	DWL (Rs.)	Cases	% of cases
Cash-only	<50	25	135	208	32.7
Single-switch					
	50–100	75	85	37	5.8
	100–150	125	35	103	16.2
	150–200	175	−15	106	16.6
	200–250	225	−65	48	7.5
	250–300	275	−115	25	3.9
	300–350	325	−165	6	0.9
	350–400	375	−215	11	1.7
	400–500	450	−290	5	0.8
Rice-only	>500	550	−390	88	13.8
Average/Total (Single-switch)		177	-17^{***}	341	53.5
Average/Total (All types)		179	-19^{***}	637	100

Notes: Willingness to pay (WTP) for rice is defined as the midpoint of the switch interval in which the respondent switched from rice to cash. DWL is calculated as $160 - \text{WTP}$, where Rs. 160 is the market value of 5 kilograms of rice. Respondents with multiple switches are excluded. WTP for rice-only respondents is assumed to be Rs. 550; for cash-only respondents, Rs. 25. ***Statistically significant at the 1% level.

It is worth noting that our overall measure of a DWG of Rs. 19 is an underestimate because we bound our cash choices at a maximum of Rs 500. At this upper bound, there are still 14% of respondent choices favoring rice. Had we gone on to offer higher cash amounts in our treatments, it would have led to even higher amounts of DWG. The lower bound of cash offers in contrast has a natural limit of zero.

4 The "Puzzle" of Deadweight Gain

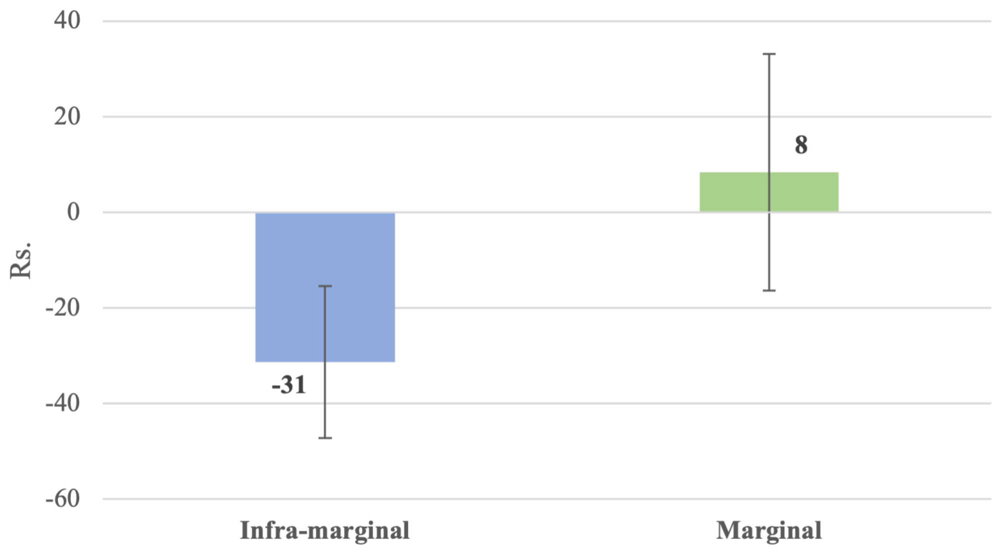
4.1 Infra-marginality

The prevalence of deadweight gain is widespread in our sample and is observed for about 45% of all cases. While this result may appear "puzzling", as discussed in the Introduction, whether beneficiaries are marginal or infra-marginal has a potentially important bearing on deadweight losses associated with in-kind transfers. We can identify these two categories of beneficiaries using baseline data on purchases of market rice. At baseline, prior to our experiment, 83% of our sample households are infra-marginal, i.e., their total rice consumption is greater than their consumption of PDS rice. Put differently, for 83% of households, the quantity of market rice consumption is greater than zero. This is consistent with the evidence on the large proportions of infra-marginal beneficiaries in many food subsidy programs where these programs have been implemented at scale, including for India's PDS program where Gadenne et al. (2024) report 93% of PDS beneficiaries as infra-marginal.

In our experiment, the offer of 5 kilograms of free rice was additional to the rice already available to the respondents through the PDS system. Therefore, *for the purposes of our experiment*, infra-marginal households can be identified as those who bought at least 5 kilograms of rice from the open market at baseline.[5] Using this definition, one-third of respondents are marginal, and two-thirds are infra-marginal (Table A2).

Based on standard theory we would have expected to see either a positive deadweight loss for marginal respondents or a zero deadweight loss for infra-marginal respondents. Splitting our sample between marginal and infra-marginal respondents, we do find a deadweight loss for marginal respondents as consistent with theory. However, for inframarginal respondents we observe a deadweight gain (Figure 4). The difference between the two groups is highly significant (p=0.007). Thus the role of infra-marginality in standard theory does not fully explain our findings.

Figure 4: Deadweight Loss for infra-marginal and marginal respondents



Note: The difference between infra-marginal and marginal respondents is significant at p=0.007.

Going beyond the standard theory, deadweight gains associated with the choice of rice could arise due to several factors, for instance, higher transaction costs incurred by respondents for the cash option relative to rice, superior quality of rice offered in the experiment, or limited trust in redeeming the cash option. An additional dimension that deserves consideration relates to the role of intra-household bargaining in shaping respondent choices between cash or rice. We discuss these potential factors below.

[5]The five kilograms of rice offered in the experiment represent approximately 30 percent of the average monthly consumption of rice for the respondents in our sample (Table A2).

4.2 Transaction costs and rice quality

The choice between cash or rice would be clearly influenced by the relative transaction costs for the respondents associated with each option. Cash transfers are typically delivered through deposits into respondent accounts with financial institutions (banks or post offices). Transaction costs for the cash option are thus determined by several factors such as the density and capacity of the financial network, the ease of operating bank accounts, and the financial literacy of respondents. Similarly, transaction costs for the rice option depend on the proximity to and familiarity with the local rice shop. In addition to transaction costs, respondent choice could also be influenced by the quality of subsidized rice, with higher willingness to pay for better quality rice. Our experiment, by design, controls for both these factors. Transaction costs for the cash and rice options are identical in our experiment since cash or rice are both delivered through vouchers redeemable at the same local shop. In our experiment, the respondents were offered vouchers for rice of a quality comparable to what the PDS provides. Thus, superior quality of rice offered cannot explain a higher willingness to pay for rice.

4.3 Trust

Lower trust in the delivery of the cash option is unlikely to explain deadweight gains in our experiment for three reasons. First, the experiment was preceded by a pilot, which was run as a practice round with the full sample of respondents. The pilot was implemented with full protocols of the experiment, and all vouchers for cash or rice issued in the pilot run were successfully redeemed. Thus, we believe that by end of the pilot the respondents trusted the implementation of the incentivization mechanism. Second, since vouchers were given for both cash or rice, any potential trust issues would be similar for both cash and rice.[6] Third, none of the respondents in the three rounds reported any concerns or difficulties with redeeming the vouchers for cash or rice at the local shop.

4.4 Intra-household bargaining

Survey-based and qualitative evidence points to the potential role of intra-household inequality and gender in influencing the choice over in-kind transfers or cash (Ghatak et al., 2016; Khera, 2011, 2014). Survey responses highlight that in-kind transfers provide some protection from intra-household disputes over cash: "*Sometimes her young children [may] sleep hungry as her husband drinks. Nobody will snatch rice*"; "*The ration we get is quite alright. Cash will be spent on alcohol, and nothing will remain for our children. If we get rice, everyone will share [eat] it*" (Khera, 2014).[7] Similarly, Drèze and Sen (2013) note that: "Cash is also more easily deflected towards the purchase of goods that are consumed mainly by adult members of the family, especially men, at the expense of undernourished girls and other children".

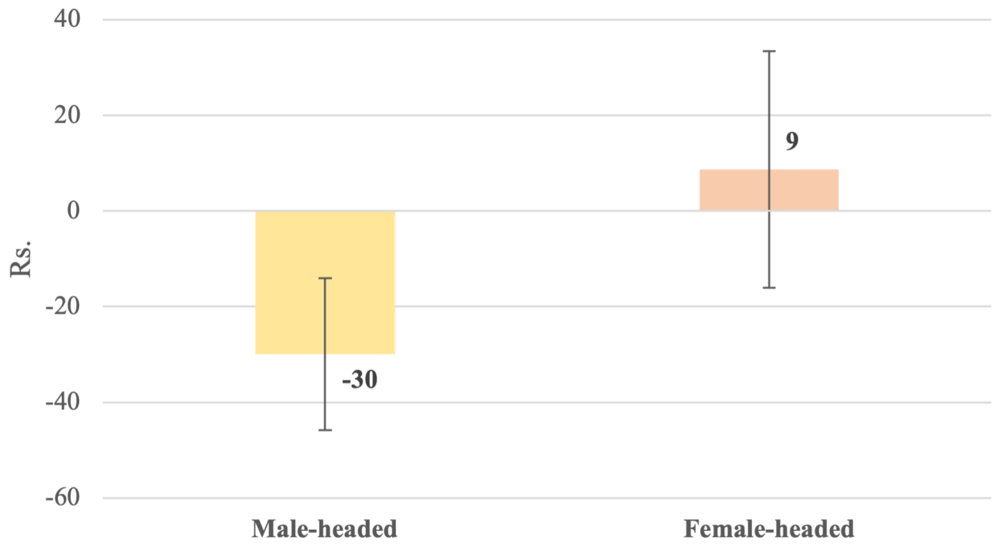
[6] Nor can such trust considerations explain the differential willingness to pay for rice between respondents from male- and female-headed households, the evidence for which is presented in section 4.4

[7] A similar sentiment was echoed in an interview with a female respondent in Khera (2011): "...even if you give me Rs 1 lakh [one hundred thousand, approximately USD 2,190, around the time of the interview], I will opt for rice."

We therefore look further into the "puzzle" of the deadweight gain by investigating if the results differ by some measure of gender differentials in control over the household budget. One such measure that we can identify in our data is the gender of the head of the household. This allows us to distinguish cases where women have control over decisions about both cash and the subsidized product from those where women's control is limited to the subsidized product.

In line with the survey-based evidence, we would expect female-headed household to favor cash over rice. This is indeed what we find. Respondents in female-headed households have a significantly lower WTP for rice than those in male-headed households (Rs 151 as against Rs 190; p-value=0.006). As a result, in contrast to a deadweight loss for female-headed households on average of Rs 9 (5% of the value of subsidized rice), we observe a deadweight gain for male-headed households of Rs 30 (19% of the value of subsidized rice); see Figure 5.

Figure 5: Deadweight Loss for respondents from male- and female-headed households



Note: The difference between male- and female-headed respondents is significant at p=0.006.

Motivated by this finding, in Appendix D, we present a conceptual framework of intra-household bargaining that can explain why individuals may make choices biased towards in-kind benefits. The intuition underlying the framework can be briefly described as follows. In the typical Indian setting of male-headed households, the woman is often in charge of managing the food supply out of a given food budget, while her husband controls the rest of household finances. An in-kind benefit of rice can benefit the woman in this setting more than a cash transfer of similar value, as she can save money out of her food budget that otherwise would have had to be spent on rice. This is the basis for expecting among women from male-headed households a higher likelihood of choosing rice relative to women from female-headed households. With the male-headed households being more ubiquitous, there is an overall observed bias towards in-kind benefits.

The next section reports our detailed results on deadweight losses or gains and the respondents' choices between cash or rice.

5 Detailed Results

5.1 Deadweight loss or gain?

Recall that the information on the switch points or the rice/cash-only choices by respondents allows us to construct measures of DWL for each respondent and each round. We then estimate the following DWL model:

$$\begin{aligned} DWL_{ihst} = \gamma^{DWL} FemaleHead_{ihs} + \eta^{DWL} Marginal_{ihs} + X_{ihs}\delta^{DWL} \\ + \theta_s^{DWL} + \mu_t^{DWL} + v_h^{DWL} + w_{ihst}^{DWL} \end{aligned} \tag{1}$$

where DWL_{ihst} is the deadweight loss for respondent i from household h in slum s in round t, $FemaleHead_{ihs}$ is a binary variable representing female headship of the household the respondent belongs to, $Marginal_{ihs}$ is a binary variable which equals 1 if the respondent belongs to a marginal household (with monthly consumption up to 5 kilograms of market rice at baseline), while X is a vector of (time-invariant) respondent and household controls at baseline. θ_s and μ_t represent slum and round effects respectively. v_h represents the random effect for each household h, and w_{ihst} is a white-noise error term. Equation (1) is estimated using the random effects estimator.[8] Our parameters of interest are those for $FemaleHead$ and $Marginal$, both of which we expect to be positive in light of the foregoing discussion and as predicted in the theoretical framework in Appendix D.

Columns (1) and (2) of Table 4 present random effects estimates of equation (1) with varying sets of controls. The controls include baseline characteristics of the respondent (age, caste and religion) and the household (proportion of literate members, dependency ratio, the count of assets, monthly per capita expenditure, house ownership, whether the household has a ration card, and whether a female redeemed the voucher). Analogous to the unconditional estimates shown in Figure 4, we find that the parameter for marginal respondents is positive and significant. This is consistent with the standard theoretical prediction that in-kind transfers would induce deadweight losses for marginal respondents. Marginality increases DWL by Rs. 45 on average, about 28% of the market value of 5 kilograms of rice (column 2 of Table 4).

Notably, female headship is also positive and significant in all specifications of DWL. We find that female headship increases DWL by Rs 41-45, or about 26-28% of the market value of 5 kilograms of rice (columns 1 and 2). Put differently, relative to female-headed households, women in male-headed households put a 26-28% premium on subsidized rice.

Table 4 also reports additional specifications which explore the robustness of these results to how the DWL measure is constructed in our experiment. First, noting that the measures of WTP and DWL are not well-defined for the cash-only and rice-only households, in column (3),

[8] The random effects estimator allows us to control for unobserved random household attributes (such as random household-level liquidity shocks) while still identifying the effect of female headship and marginality. Note that the ordinary least squares estimates with standard errors clustered at the household level are very similar to the random effects estimates.

we set the WTP for rice for the former at zero, and that for the latter at Rs 650 (about four times the market value of 5 kilograms rice). Second, Table 4 also shows specifications where we set WTP using the bounds that are either most favorable to cash (i.e. lower bounds of WTP for rice) or most favorable to rice (i.e. upper bounds of WTP for rice) in columns (4) and (5) respectively. In all these specifications, columns (3)-(5), female headship continues to be positive and highly significant. The point estimates are similar to those in columns (1) and (2), indicating a 23-30% premium on rice among respondents in male-headed households.[9]

Table 4: Regressions of deadweight loss

	Dependent variable: DWL				
	Random effects (RE)		Robustness checks		
			RE with different bounds for willingness to pay (WTP) for rice		
	WTP=25 for rice only, WTP=550 for cash only, WTP=mid-point of switch interval for others		WTP=0 for rice only, WTP=650 for cash only, WTP=mid-point of switch interval	WTP=lower bound of switch interval	WTP=upper bound of switch interval
	(1)	**(2)**	**(3)**	**(4)**	**(5)**
Female head	44.79**	41.06*	48.24*	40.01**	36.87*
	(19.88)	(21.05)	(25.66)	(20.18)	(19.20)
Marginal respondent	47.53**	44.74**	56.70**	41.96**	40.96**
	(19.48)	(21.72)	(26.53)	(20.82)	(19.79)
Round 2	28.31*	26.45*	29.62	26.17*	24.79*
	(15.21)	(15.48)	(18.93)	(14.80)	(14.18)
Round 3	66.59***	63.50***	74.61***	61.12***	60.43***
	(14.79)	(14.87)	(18.14)	(14.26)	(13.58)
Constant	-102.58***	-150.96**	-176.22**	-172.97***	-113.26*
	(19.86)	(66.24)	(80.14)	(63.85)	(60.58)
Respondent/household controls	No	Yes	Yes	Yes	Yes
Slum effects	Yes	Yes	Yes	Yes	Yes
Mean of dep. variable	-18.65	-18.07	-23.57	-39.84	2.37
N	637	636	636	636	636
R^2	0.10	0.11	0.11	0.11	0.11

Notes: * p<0.10, ** p<0.05. *** p<0.01. "Marginal respondent" refers to respondents with purchase of up to 5 kilograms of market rice in the previous month. Standard errors in parentheses are clustered at the household level. Respondent/household-level controls include: age of the respondent, binary variable =1 if the respondent is Hindu, binary variable =1 if the respondent belongs to a scheduled caste or scheduled tribe, proportion of literate members in the household, dependency ratio, the count of household assets, baseline monthly per capita expenditure of the household, binary variable =1 if the household owns their house, binary variable =1 if the household has a ration card, and binary variable =1 if a female redeemed the voucher.

These results suggest two important explanations for the existence of deadweight losses associated with in-kind transfers: marginality of transfers for some respondents, and the role of women's bargaining power in how respondents choose between cash and in-kind transfers. While the role of marginality is as expected from standard theory, the significance of female headship while controlling for marginality is an important finding. It provides evidence for intra-household bargaining considerations highlighted in the literature on collective models of household behavior. Collective models of household behavior have however seldom been applied to the issue of choice between cash or in-kind transfers. Our finding on the role of female headship as a "distribution factor" determining women's relative bargaining power within the household is significant in this context (Browning et al., 2006, 2013). Distribution factors in this literature refer to extra-environmental parameters which influence the Pareto weights on

[9]Recognizing the discrete nature of all the cash options in our experiment that allow us to observe only an interval for WTP and DWL, we also estimated equation (1) using a random effects interval regression estimator (McDonald et al., 2018; Stewart, 1983). Female headship continued to be positive and highly significant. Similarly, female headship was also positive and significant when treating DWL as a binary dependent variable (1 for positive DWL, 0 otherwise). Results from these specifications are available upon request.

the individual (male/female) utility functions. In empirical applications, these have included factors such as sex ratio in the surrounding population, wealth contributed by each member at marriage, and the level of single parent benefit. Analogously, in our study, female headship can be viewed as a distribution factor.

We also compare the deadweight loss for male-headed and female-headed households restricting the latter to those where female heads were widowed or separated or had an absentee husband.[10] Clearly, we can expect these female heads to be in charge of the household budgets. In line with this expectation, Appendix Table C1 shows that the DWL for this set of female-headed households exceeds that for male headed households by a magnitude similar to that in the case of the full set of female-headed households (as in Column 2 of Table 4). This is consistent with the interpretation of our results in terms of intra-household bargaining power.

5.2 Cash-or-rice?

We further investigate the respondent choices underlying the deadweight losses or gain by utilizing the dataset for all the nine cash options offered to the respondents as an alternative to the in-kind offer of 5 kilograms of rice. We estimate the following model:

$$\begin{aligned} Y_{cihst} = {} & \beta_c + \gamma FemaleHead_{ihs} + \eta Marginal_{ihs} + X_{ihs}\delta \\ & + \theta_s + \mu_t + v_h + w_{cihst} \end{aligned} \tag{2}$$

where subscript c denotes the cash option, i denotes the respondent from household h, s denotes the slum and t denotes the round of the experiment. Y is a binary variable which equals 1 if the respondent chose cash instead of rice, 0 otherwise. β_c are the parameters for the nine cash options representing the marginal effects on the probability of choosing cash as the amount of cash offer increases. $FemaleHead$ and $Marginal$ are defined as before representing binary variable for female headship and marginality. X, θ_s, μ_t, v_h and w_{cihst} are also defined as in the deadweight loss equation (1) above. Equation (2) is estimated using the random effects estimator. Our parameters of interest are β_c, γ and η. We expect β_c to be increasing in the cash amounts offered as the rice option becomes progressively less attractive. The parameter on marginality, η, is expected to be positive since marginal respondents are predicted to have higher propensity to choose cash. Further, in line with the foregoing discussion on intra-household bargaining, we expect respondents from female-headed households to have a higher probability of choosing cash (i.e., $\gamma > 0$).

Table 5 report the estimates of equation (2). Column (1) presents estimates of β_c without

[10]Female-headed households comprise 29% of our sample, of which 73% (19% of the total sample) reported the husband either dead, separated or not present in the household. The rest of the female-headed households report their marital status as currently married with a living husband, but in these cases the husband is either away (e.g. working on construction sites or as security guard) or is unable to work (likely due to disability). For the state of Maharashtra as a whole, the estimated proportion of female-headed households is 15% (the National Survey Sample Organization's Time Use Survey 2019). The higher proportion of female-headed households in our sample reflects the relatively higher proportion of absentee or non-working husbands in the low-income urban neighborhoods which formed our sample frame.

any covariates, and essentially reproduces the findings in Table 1. Consistent with expectation, relative to the reference cash offer of Rs 50, all β_c's are positive, significant, and increase monotonically as the cash offer increases. Columns (2) and (3) introduce female headship and marginality, while column (3) also introduces additional controls for baseline characteristics of the respondent and the household (same as those for the DWL regressions).

Table 5: Random effects linear probability model of choice between cash and rice options

Dependent Variable: 1 if respondent chose cash, 0 if they chose rice	(1)	(2)	(3)
Cash 100	0.06***	0.06***	0.06***
	(0.01)	(0.01)	(0.01)
Cash 150	0.22***	0.22***	0.22***
	(0.02)	(0.02)	(0.02)
Cash 200	0.39***	0.39***	0.39***
	(0.02)	(0.02)	(0.02)
Cash 250	0.46***	0.46***	0.46***
	(0.02)	(0.02)	(0.02)
Cash 300	0.50***	0.50***	0.50***
	(0.02)	(0.02)	(0.03)
Cash 350	0.51***	0.51***	0.51***
	(0.02)	(0.02)	(0.02)
Cash 400	0.53***	0.53***	0.53***
	(0.03)	(0.03)	(0.03)
Cash 500	0.53***	0.53***	0.53***
	(0.03)	(0.03)	(0.03)
Female head		0.10**	0.09**
		(0.04)	(0.04)
Marginal respondent		0.09**	0.08*
		(0.04)	(0.04)
Round 2	0.06**	0.06**	0.06**
	(0.03)	(0.03)	(0.03)
Round 3	0.13***	0.13***	0.12***
	(0.03)	(0.03)	(0.03)
Constant	0.17***	0.15***	0.03
	(0.04)	(0.04)	(0.14)
Slum effects	Yes	Yes	Yes
Respondent/household-level controls	No	No	Yes
N	5733	5733	5724
R^2	0.21	0.23	0.23

Notes: * p<0.10, ** p<0.05, *** p<0.01. Standard errors in parentheses are clustered at the household level. Respondent/household-level controls include: age of the respondent, binary variable =1 if the respondent is Hindu, binary variable =1 if the respondent belongs to a scheduled caste or scheduled tribe, proportion of literate members in the household, dependency ratio, the count of household assets, baseline monthly per capita expenditure of the household, binary variable =1 if the household owns their house, binary variable =1 if the household has a ration card, and binary variable =1 if a female redeemed the voucher.

In all specifications, we also control for round effects, and find that the probability of choosing cash increases in later rounds. This is consistent with survey-based evidence of an increasing preference for cash over time observed by Muralidharan et al. (2017) for a pilot program, which introduced cash transfers in lieu of food rations, in three Union Territories of India. They find that while initially 39% of beneficiaries preferred cash over food, by the end of their year-long survey this rose to 65%.[11] The variation across rounds is discussed further in section 7.1. Consistent with predictions from standard theory, we find that marginality increases the probability of choosing cash by 8-9 percentage points. Specifically, η is positive and significant in columns (2) and (3).

Our key result relates to the parameter for female headship (γ), which we find to be positive and statistically significant while controlling for marginality. Female headship increases the probability of choosing cash by 9-10 percentage points.[12] Insofar as women's bargaining power is likely to be higher in female-headed households relative to male-headed households, this result is in line with our conceptual framework (Appendix D), which predicts that with a greater control of the household budget, women are more likely to prefer cash.

This finding on the significance of female headship for the choice between cash or rice also suggests the possibility that the marginal effect of female headship varies by the amount of cash offered. We investigate this by interacting cash options with female headship as in the following regression:

$$\begin{aligned} Y_{cihst} &= \beta_c^F FemaleHead_{ihs} + \beta_c^M MaleHead_{ihs} + \eta Marginal_{ihs} \\ &\quad + X_{ihs}\delta + \theta_s + \mu_t + v_h + w_{cihst} \end{aligned} \tag{3}$$

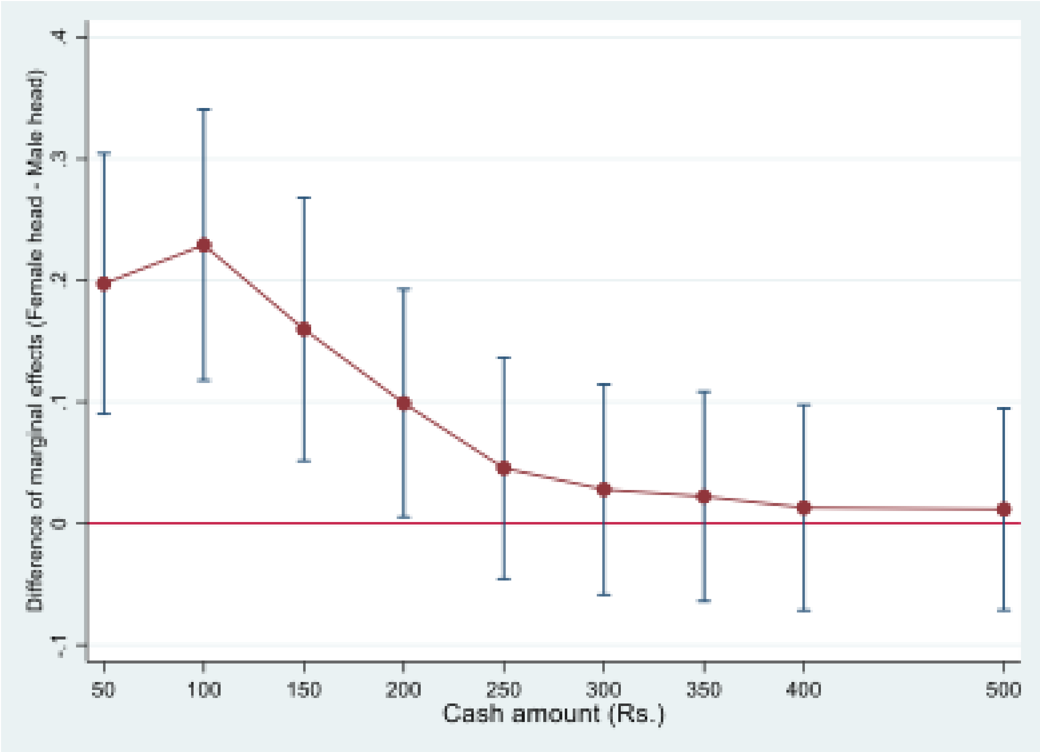
We find that while both β_c^F and β_c^M are increasing in the cash amount, β_c^F for female-headed households is higher than β_c^M for male-headed households at every cash option. Figure 6 shows the results for the difference in the marginal effects ($\beta_c^F - \beta_c^M$). The difference is larger at lower cash offers and gradually reduces as cash offers increase, with the marginal effects eventually converging across male and female-headed households. For instance, the difference in marginal effects is about 21-24 percentage points at cash amounts of Rs 50-Rs 100, but reduces to 10 percentage point at the cash offer of Rs. 200 and falls further down to 2 percentage points at Rs 400-Rs 500. Testing for statistical significance, we find that the marginal effects for female-headed households remain significantly higher up to a cash amount of Rs 200, but not thereafter.

Our main findings thus indicate that women in male-headed households (in light of their lower bargaining power) are more likely to choose rice than women in female-headed households so long as the difference between the market value of rice and the cash offer is not too large. When subject to conditions of lower bargaining power, women are willing to forgo a certain amount of cash as a strategy to protect their share of the household budget. This is an important basis of the deadweight gain we observe in our experiment.

[11] In related work, Hidrobo et al. (2014) report that respondents show stronger preferences for the type of transfer they are familiar with.

[12] The additional controls introduced in column (3) turn out to be insignificant, and our parameters of interest remain unchanged both in magnitude and significance.

Figure 6: Difference in the marginal effects of the cash amount on the probability of choosing cash (Female-headed *minus* Male-headed)



Note: Based on parameter estimates of equation (3) where the cash offer variable is interacted with female and male headship.

6 Rice as a commitment device?

Another potential reason for observing deadweight gains in our data could be that respondents want to use the rice option as a commitment device. A commitment device is understood in the literature as an arrangement used by an individual with the aim of helping fulfill a plan for future behavior that would otherwise be difficult to achieve (Bryan et al., 2010). This could be due to intrapersonal conflict such as limited self-control, or due to strategic motives related to interpersonal influences on own behavior.

In our experiment, we collected data which allows us to examine the possible use of rice as a commitment device. Once the choices for cash or rice were made against the nine cash offers and one of these was randomly selected through the lottery for implementation, we asked the respondents the primary reason for their choice in the implemented option. The possible reasons for choosing rice included: (i) I chose rice because the cash amount is less than the value of 5 kilograms of rice, (ii) I chose rice because cash will get spent on less useful things than rice, (iii) I chose rice because we are running short of rice, (iv) I chose rice because it is hard to control how cash will get spent, and (v) others.[13] In our sample, the frequencies of these reasons were 27, 27, 30, 14 and 2 percent respectively.

Using this data, we define a binary variable, Rice as Commitment Device, which equals 1 if

[13]The possible reasons for the cash choice were also elicited, and are shown in Appendix E.

the respondent reported (ii) or (iv) as the primary reason for their choice of rice (0 otherwise). In our sample, about 40 percent of respondents indicate commitment considerations for their choice of rice. It is of interest to enquire whether such commitment considerations are also related to female headship. Columns (1) and (2) of Table 6 show that respondents from female-headed households are less likely to report commitment considerations for their choice of rice. This also holds when we further control in Columns (3) and (4) for the lottery amount being less than Rs. 200 (in which case respondents would be more likely to choose rice because the value of rice is greater than the cash offer).[14] In other words, this result suggests that while commitment considerations may be relevant for all respondents perhaps for self-control reasons, women from male-headed households feel more of a need to use rice as a commitment device relative to their counterparts from female-headed households, which suggests that intrahousehold strategic motives are also in play. Given that there are no significant socioeconomic differences between male- and female-headed households (as we document in section 7.2), this is indicative of the underlying lower bargaining power of women in male-headed households.

Table 6: Rice as commitment device

	Dependent variable: Rice as Commitment Device (binary)			
	(1)	(2)	(3)	(4)
Female head	-0.062**	-0.066**	-0.061**	-0.064**
	(0.025)	(0.027)	(0.025)	(0.027)
Lottery amount			-0.100***	-0.097***
(=1 if lottery amount < 200)			(0.032)	(0.032)
Respondent/household controls	No	Yes	No	Yes
Slum effects	Yes	Yes	Yes	Yes
N	637	636	637	636
R^2	0.06	0.08	0.08	0.10
Mean of dep. variable	0.115	0.115	0.115	0.115

Notes: * p<0.10, ** p<0.05, *** p<0.01. Standard errors in parentheses are clustered at the household level. Respondent/household-level controls include: age of the respondent, binary variable =1 if the respondent is Hindu, binary variable =1 if the respondent belongs to a scheduled caste or scheduled tribe, proportion of literate members in the household, dependency ratio, the count of household assets, baseline monthly per capita expenditure of the household, binary variable =1 if the household owns their house, binary variable =1 if the household has a ration card, and binary variable =1 if a female redeemed the voucher.

7 Alternative Explanations

The foregoing discussion highlights three explanations for the observed deadweight gains or losses in our sample: marginality or infra-marginality of in-kind transfers; possible use of rice as a commitment device; and intra-household differentials in bargaining power. In this section, we discuss further alternative explanations for our findings, specifically, those related to (i) rice as

[14] It is possible that the third reason for the choice of rice, viz. "I chose rice because we are running short of rice" could also reflect commitment concerns related to ensuring food security for the household, as cash could be diverted to other purchases. Upon including this reason in our definition of the commitment device variable, we find that female headship remains negative and significant.

insurance against price volatility, and (ii) and whether learning or renegotiation could explain respondent choices over rounds. The discussion for (ii) also draws upon the evidence on why female headship does not represent socioeconomic differences, but instead is a plausible proxy for women's bargaining power.

7.1 In-kind transfer as insurance against price risk

In an environment of price volatility, in-kind transfers can confer insurance benefits to recipients in addition to their pure redistributive value. Gadenne et al. (2024) show that if the marginal utility of income is increasing in the price of the subsidized product, inframarginal in-kind transfers may be preferred by recipients over equivalent cash transfers for their insurance value since the value of the transfer automatically increases with the price of the transferred product. We are unable to explore this mechanism in our setting as the three rounds of our experiment are limited to a five-month period over which there is little temporal variation in the price of rice (see section 2). In addition, the respondents in our experiment were not informed that there will be three rounds of the experiment. From their perspective, each round was effectively the last round. Therefore, within this experimental setup, price risk considerations, which presuppose the continuation of the intervention, are not likely to be relevant.

Further, insurance against price risk is an unlikely mechanism driving our main results for two additional reasons. First, while we observe, as noted above, a lower likelihood of choosing rice in round 3, there is no evidence of a lower price of rice in August 2019 when round 3 was conducted. Second, price variability cannot explain our main finding of differential responses across male- and female-headed households who face the same set of prices. Indeed, the differential response to similar prices is consistent with our main explanation relating to the role of intra-household factors. Our estimates nonetheless control for round effects, making them robust to any temporal price variations that may exist.

7.2 Learning or renegotiation?

As noted above (sections 3 and 5), the respondents' choice in favor of rice relative to cash declines in round 3 of the experiment, implying lower deadweight gains in round 3. This finding seems consistent with the notion that respondents learn over time that cash offers above the market value of 5 kilograms of rice are more attractive. We explore this further by estimating equation (1) separately by round as shown in Table 7.

We find that female headship continues to be positive and significant for round 2 and for rounds 1 and 2 combined. However, the marginal effect of female headship is much lower in round 3 and is not statistically significant. This could be due to a change in respondent choices in either male- or female-headed households or both. To investigate this further, we delve into the respondents' cash or rice choices underlying the deadweight loss regression.

Specifically, we test for the stability of marginal effects on the probability of choosing cash across the three rounds separately for female- and male-headed households. The results

Table 7: Regressions of deadweight loss by survey round

	(1)	(2)	(3)	(4)
	Round 1	Round 2	Rounds 1 & 2	Round 3
	DWL			
Female head	40.33	61.08**	49.52**	16.53
	(31.44)	(29.95)	(22.02)	(25.61)
Respondent/household-level controls	Yes	Yes	Yes	Yes
Slum effects	Yes	Yes	Yes	Yes
N	205	220	425	211
Mean of dep. variable	-46.10	-25.80	-35.59	17.23

Notes: * p<0.10, ** p<0.05, *** p<0.01. Standard errors in parentheses are clustered at the household level. Respondent/household-level controls include: binary variable for marginal respondent, age of the respondent, binary variable =1 if the respondent is Hindu, binary variable =1 if the respondent belongs to a scheduled caste or scheduled tribe, proportion of literate members in the household, dependency ratio, the count of household assets, baseline monthly per capita expenditure of the household, binary variable =1 if the household owns their house, binary variable =1 if the household has a ration card, and binary variable =1 if a female redeemed the voucher.

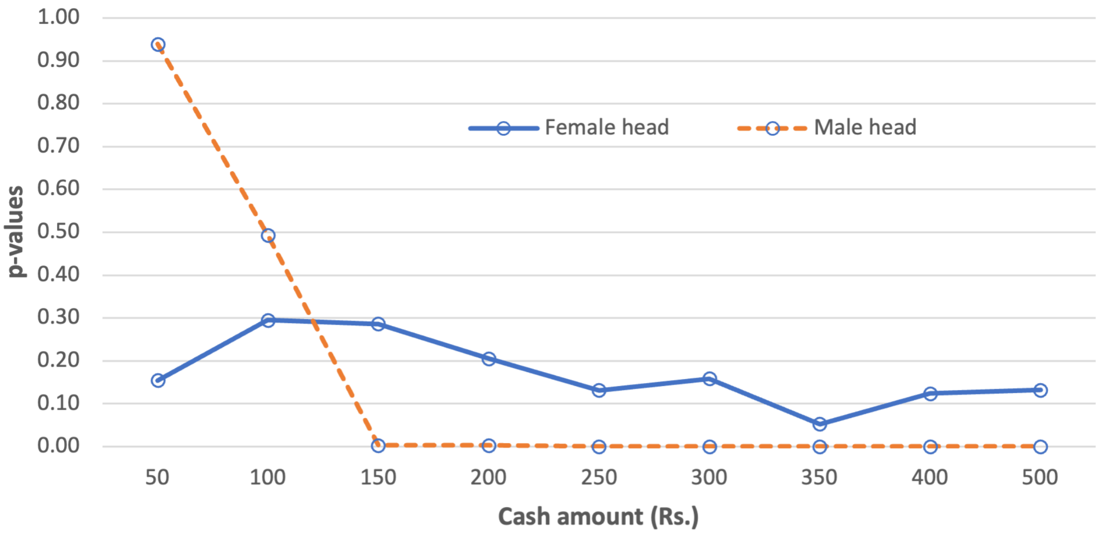
of these tests are shown in Figure 7. For female-headed households, the marginal effects are not statistically different across rounds for all cash offers (as indicated by the p-values higher than 10%) with the exception of cash 350. In contrast, the marginal effects for male-headed households are statistically different across rounds for all cash offers except cash 50 and cash 100. In other words, it is women from male-headed households that exhibit a change in behavior in round 3. Thus, the lower (and insignificant) effect of female headship on deadweight loss in round 3 (Table 7) is attributable to a change in choices by women not from female-headed households, but to those from male-headed households.

To interpret these results in terms of learning over time, one would have to assume differential rates of learning across women from male- and female-headed households. However, there is no a priori reason to expect this given that, for our sample many socioeconomic characteristics are similar for the two types of households (see Table 8, top panel). For instance, there are no significant differences by male or female headship with respect to the proportion of literate members, dependency ratio, per capita monthly expenditure, share of food expenditure, asset count and house ownership.[15] On the other hand, our data indicate systematic differences by male or female headship in a number of decision-making and intra-household bargaining variables (see Table 8, bottom panel). Female headship is associated with significantly higher proportion of decisions made by women.[16] Women in female-headed households are also significantly more likely to buy market grain, use a bank account and be the food supply manager. Thus, for our

[15] Similarly, data from the NSS Time Use Survey of 2019 indicates that female-headed households are not necessarily poorer; for instance, while the proportion of female-headed households for the bottom two quintiles of per capita expenditure is 14%, it is 17% for the top two quintiles. For similar evidence on female headship being a poor indicator of socioeconomic status, see Brown et al. (2019).

[16] The baseline questionnaire specifically asked about whether the respondent or her spouse made the final decision on 16 major activities such as children's schooling, use of contraception, spending on food and other major household items, selling assets, taking loans, migration for employment and obtaining healthcare for children or self during the last 12 months. The proportion of decisions made by women is calculated as the fraction of cases where the final decision on an activity was made by the female respondent.

Figure 7: Tests of equality of marginal effects across rounds by cash amount



Notes: The figure plots the p-values for tests of the null hypothesis that the marginal effects on the probability of choosing cash are equal across the three rounds. The tests are done separately for female- and male-headed households. P-values below 0.05 (0.10) indicate rejection of equality at the 5% (10%) level of significance.

sample female headship does not seem to represent adverse socio-economic circumstances of the household which may induce differential learning; instead, it is indicative of women's greater bargaining power.[17]

In light of this, a plausible explanation for the lower deadweight gains in round 3 could be that in male-headed households there is a renegotiation of the food budget allocated to the woman. The renegotiation occurs in response to the fact that the first two rounds of the experiment involved provision of free rice or a cash transfer. One can expect that given this "windfall gain" to women, the male head would reduce the woman's budget allocation by an amount less than or equal to the value of the rice or cash received. This would lead to a reduced bargaining premium to choosing rice for women in these households over time. In female-headed households where women have greater if not full control of the budget, there is no similar impetus for renegotiation.

8 Conclusion

Despite a significant interest amongst researchers and policymakers in understanding the relative merits of cash vs in-kind transfers, evidence on beneficiary choices between these options and their underlying drivers remains scant. Our study investigates this question by designing and implementing an incentivized experiment in the context of the world's largest food subsidy

[17]From our theoretical framework outlined in Appendix D, it is possible to infer the implicit bargaining power of women ($\alpha(b)$) from the revealed switch points from rice to cash, as the standardized ratio of the market value of rice to WTP (i.e., 160/WTP). For single-switch households, the average value of $\alpha(b)$ is 0.4. As expected, the average $\alpha(b)$ for female-headed households (0.47) is significantly higher than that for male-headed households (0.38) with a p-value of 0.009 for the difference.

Table 8: Are female-headed households different from male-headed households?

	Female	Male	Difference	Std. error	N
Socioeconomic variables:					
Proportion of literate members	0.73	0.69	0.04	0.03	250
Dependency ratio	0.66	0.70	-0.05	0.09	249
Monthly per capita expenditure	1971.14	1917.70	53.44	109.20	250
Share of food in total expenditure	0.82	0.83	-0.01	0.01	250
Asset count	7.29	7.04	0.24	0.36	250
Own house	0.80	0.85	-0.05	0.05	250
Religion: Hindu	0.62	0.66	-0.04	0.07	250
Have a ration card	0.83	0.82	0.02	0.06	250
Decision-making/bargaining power variables:					
Proportion of decisions made by women	0.66	0.41	0.25***	0.07	248
Female buys market grain	0.94	0.67	0.27***	0.06	250
Female uses bank account	0.67	0.17	0.50***	0.06	233
Female is food supplies manager	0.95	0.40	0.55***	0.06	233

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The coefficients and standard errors are based on regressions of each socioeconomic and decision-making/bargaining power variable on female headship.

program in India. The experiment offered respondents, predominantly women in our setting, an incentivized choice between varying amounts of cash and a fixed quantity of rice. The revealed choices are then used to construct estimates of the recipients' willingness to pay for rice and hence the associated deadweight loss.

We find evidence of deadweight gain on average in our experimental data. Our analysis points to three lines of explanation. First, consistent with standard theory, we find that marginality contributes to higher deadweight losses associated with in-kind transfers. Second, controlling for marginality our analysis reveals a striking contrast between respondents from male- and female-headed households, which reflects the underlying role of gender differences in bargaining power in influencing respondent choices. We find that the observed overall deadweight gain is a consequence of a deadweight loss among respondents in female-headed households and a deadweight gain among those in male-headed households. Given that most households are male-headed, the deadweight gain dominates. Third, our results indicate that behavioral factors play a role in inducing deadweight gains. In particular, we find evidence for the respondents' choice of in-kind transfers as a commitment device. Many respondents in both male- and female-headed households report commitment considerations as their reason for choosing rice over cash. However, these considerations seem to extend beyond issues of self-control, and also reflect behavioral actions by respondents to limit others' control over their choices. In support of this interpretation, women in male-headed households are significantly more likely to report commitment motivations than their counterparts in female-headed households. This finding underscores that intra-household bargaining consideration may also be relevant to the use of in-kind transfer as commitment devices.

Most welfare programs are designed to provide either only cash or only in-kind transfers. The existence of deadweight gains associated with in-kind transfers in our experiment does not necessarily imply that in-kind transfers are the preferred policy option. Rather, a key policy insight of our study is that there is a case for offering respondents a choice between cash or kind. Such flexibility in program design can accommodate different incentives faced by marginal and infra-marginal households, allow the use of in-kind transfers as commitment devices as needed, and can be an important instrument for those with weaker bargaining power to sustain a measure of control over the household budget.

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Appendix

A Survey Details

This Appendix presents (i) the details of the listing operation conducted to select respondents for the experiment, (ii) photos of vouchers used in the experiment, and (iii) summary statistics for our sample.

A.1 Details of the listing operation

The listing operation collected data on the address of household, name of the household head, family size, whether the household has a ration card and how much rice (in kilograms) the household purchased from the market in a month?

Table A1: Listing Operation Sample

Sl. No.	Slum Name	Household Population	Households Listed
1	Lekha Nagar (CIDCO)	307	81
2	Indira Gandhi Nagar (Upnagar, Nasik Road)	457	103
3	Sant Kabir Nagar (Dwarka Poona Road)	252	102
4	Rahul Nagar (Golfclub, West)	133	93
5	Sahjeevan Nagar (Ganesh Wadi, Panchavati)	147	111
6	P. C. Tolls Prabudha Nagar (Mahindra Front Satpur)	1333	100
7	Sant Kabir Nagar (Canal Satpur Bhosala)	767	99
8	Kolivada (Nashik East)	208	101
9	Mahatma Phule Nagar (Peth Road, Panchavati)	1985	101
10	Wadarvadi, Nagar (Phule Nagar)	155	100
	Total	**5744**	**991**

A.2 Vouchers

Figure A1: Vouchers for cash and rice and the surveyors interviewing households

A.3 Summary statistics

Table A2: Summary Statistics for Sample Households

Variable	Overall (N = 723)		Female Respondents (N = 637)	
	Mean	SD	Mean	SD
Female respondent	0.88	0.32	1.00	0.00
Age of the respondent	37.18	12.11	36.91	12.23
Female head	0.27	0.44	0.29	0.46
Marginal respondent[a]	0.33	0.47	0.32	0.47
Total household rice consumption/month (kg)	17.23	16.04	18.01	16.85
PDS rice consumption/month (kg)	4.98	4.42	5.06	4.40
Market rice consumption/month (kg)	12.23	15.67	12.93	16.44
Proportion of literate members	0.70	0.20	0.70	0.20
Household size	5.24	2.49	5.34	2.59
Dependency ratio	0.69	0.62	0.71	0.63
Social group: Scheduled Castes	0.51	0.50	0.49	0.50
Social group: Scheduled Tribes	0.13	0.34	0.14	0.34
Social group: Other Backward Castes	0.31	0.46	0.33	0.47
Social group: General	0.05	0.21	0.05	0.22
Religion: Hindu	0.65	0.48	0.67	0.47
Owned house	0.84	0.36	0.85	0.36
Ration card	0.82	0.38	0.84	0.37
Asset count	7.09	2.50	7.13	2.58
Monthly per capita consumption (Rs)	1932.71	763.85	1963.63	763.28
Female respondent used the voucher	0.59	0.49	0.67	0.47

[a] Equal to 1 if monthly market rice purchases were less than or equal to 5 kg.

B Results for the Sample Including Both Female and Male Respondents

The main paper presents results for the sample of female respondents, who comprise nearly 90% of all respondents who participated in the experiment. This Appendix presents corresponding results for the full sample. The tables and figures in this Appendix confirm the robustness of our results for the sample including both female and male respondents.

Table B1: Percentage of Households Choosing Rice Against Each Cash Offer, Pooled Across Rounds

Cash Offer (Rs)	50	100	150	200	250	300	350	400	500
% Choosing Rice	68.0	62.3	47.7	30.0	23.5	19.8	18.8	16.8	16.4

Figure B1: Percentage of households choosing cash or rice against each cash amount across the three rounds (corresponds to Figure 3)

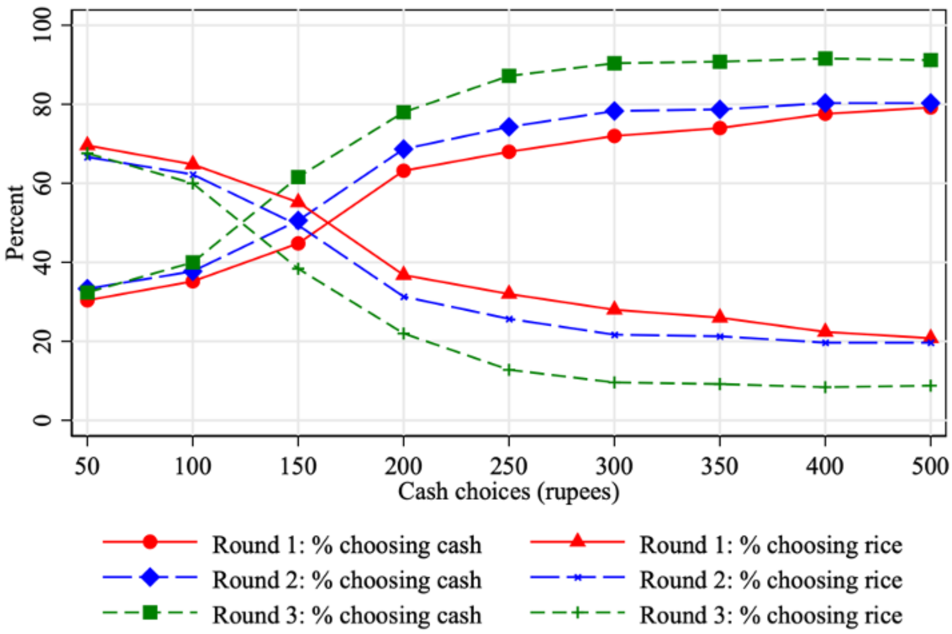


Table B2: Number and Percentage of Cases for Each Type of Household, by Round

Respondent Type	Single-switch	Rice-only	Cash-only	>1 Switch	Total
Round 1					
Number	113	45	74	18	250
% of sample*	45.2	18.0	29.6	7.2	100
% excluding single-switch	–	48.7	19.4	31.9	100
Round 2					
Number	117	46	83	3	249
% of sample*	47.0	18.5	33.3	1.2	100
% excluding single-switch	–	47.6	18.7	33.7	100
Round 3					
Number	147	19	79	5	250
% of sample*	58.8	7.6	31.6	2.0	100
% excluding single-switch	–	60.0	7.8	32.2	100
Combined					
Number	377	110	236	26	749
% of sample*	50.3	14.7	31.5	3.5	100
% excluding single-switch	–	52.1	15.2	32.6	100

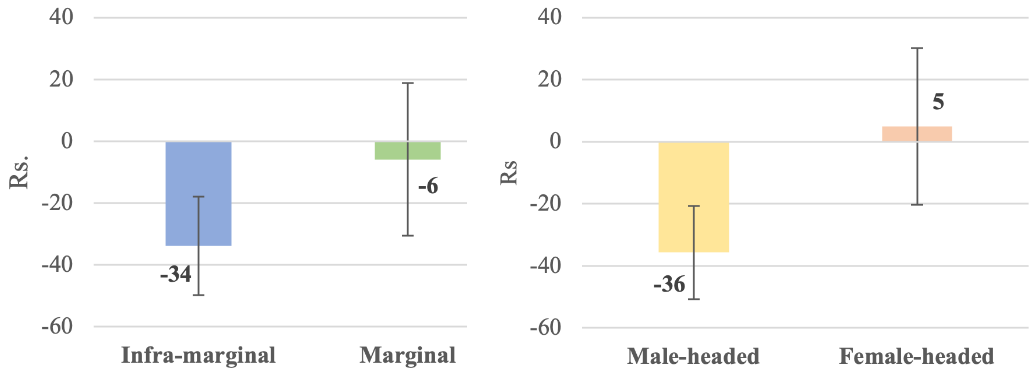
* Includes respondents with multiple switches. Final sample used in estimation excludes multiple-switch respondents.

Table B3: Willingness to Pay (WTP) and Deadweight Loss (DWL) by Household Type

Household Type	Switch Interval (Rs)	WTP (Rs)	DWL (Rs)	Number of Cases	% of Cases
Cash-only	<50	25	135	236	32.6
Single-switch	50–100	75	85	40	5.5
	100–150	125	35	110	15.2
	150–200	175	-15	121	16.7
	200–250	225	-65	51	7.1
	250–300	275	-115	28	3.9
	300–350	325	-165	10	1.4
	350–400	375	-215	12	1.7
	400–500	450	-290	5	0.7
Rice-only	>500	550	-390	110	15.2
Average/Total for single-switch		178	-18	377	52.2
Average/Total for all types		185	-25	723	100.0

Notes: Willingness to Pay (WTP) is defined as the midpoint of the cash choice interval where the respondent switches from rice to cash. DWL is calculated as $160 - \text{WTP}$, where Rs 160 is the market value of 5kg of rice. Multiple-switch respondents are excluded. Rice-only households are assigned a WTP of Rs 550; cash-only households are assigned Rs 25.

Figure B2: Deadweight Loss (DWL) in Rs (corresponds to Figures 4 and 5)



Note: The difference between infra-marginal and marginal respondents is significant at p=0.047, and between male- and female-headed respondents is significant at p=0.006.

Table B4: Regressions of deadweight loss (corresponds to Table 4)

	Dependent variable: DWL				
	Random effects (RE)		Robustness checks		
			RE with different bounds for willingness to pay (WTP) for rice		
	WTP=25 for rice only, WTP=550 for cash only, WTP=mid-point of switch interval for others		WTP=0 for rice only, WTP=650 for cash only, WTP=mid-point of switch interval	WTP=lower bound of switch interval	WTP=upper bound of switch interval
	(1)	**(2)**	**(3)**	**(4)**	**(5)**
Female head	43.68**	39.41*	46.02*	38.43*	35.55*
	(19.78)	(20.68)	(25.22)	(19.83)	(18.84)
Marginal respondent	39.78**	39.41*	49.55*	37.08*	36.12*
	(19.62)	(21.24)	(25.94)	(20.35)	(19.36)
Round 2	19.46	19.92	21.66	19.98	18.63
	(14.03)	(14.36)	(17.52)	(13.75)	(13.14)
Round 3	65.28***	64.22***	75.53***	61.83***	60.90***
	(13.83)	(14.05)	(17.14)	(13.49)	(12.81)
Constant	-95.99***	-138.36**	-159.07*	-161.77**	-100.85
	(19.00)	(67.58)	(82.10)	(64.95)	(61.80)
Respondent/HH controls	No	Yes	Yes	Yes	Yes
Slum effects	Yes	Yes	Yes	Yes	Yes
N	723	720	720	720	720
R^2	0.08	0.10	0.10	0.10	0.10

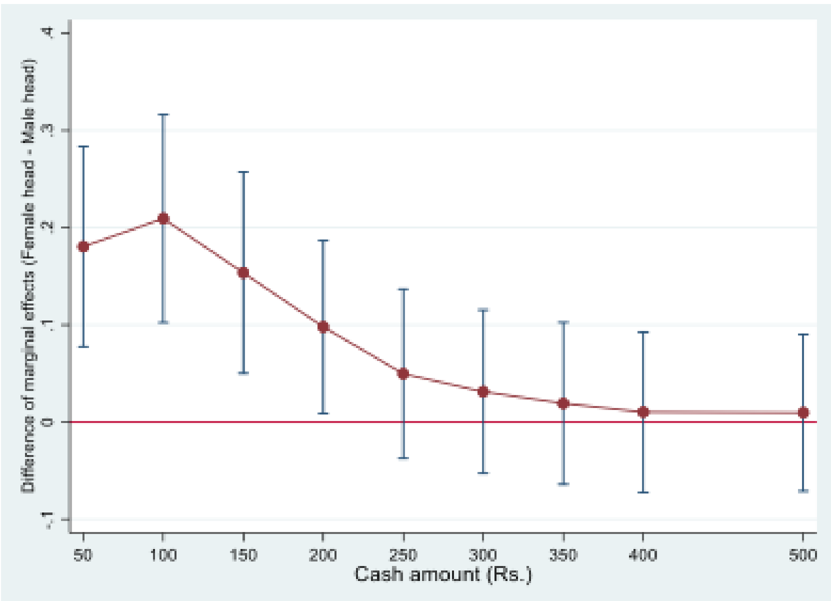
Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. "Marginal respondent" refers to respondents with purchase of up to 5 kilograms of market rice in the previous month. Standard errors in parentheses are clustered at the household level. Estimates from the overall sample including male respondents. Respondent/household-level controls include: age of the respondent, binary variable =1 if the respondent is Hindu, binary variable =1 if the respondent belongs to a scheduled caste or scheduled tribe, proportion of literate members in the household, dependency ratio, the count of household assets, baseline monthly per capita expenditure of the household, binary variable =1 if the household owns their house, binary variable =1 if the household has a ration card, and binary variable =1 if a female redeemed the voucher.

Table B5: Random effects linear probability model of choice between cash and rice options (corresponds to Table 5)

Dependent Variable: 1 = Cash, 0 = Rice	(1)	(2)	(3)
Cash 100	0.06***	0.06***	0.06***
	(0.01)	(0.01)	(0.01)
Cash 150	0.21***	0.21***	0.21***
	(0.02)	(0.02)	(0.02)
Cash 200	0.37***	0.37***	0.38***
	(0.02)	(0.02)	(0.02)
Cash 250	0.45***	0.45***	0.45***
	(0.02)	(0.02)	(0.02)
Cash 300	0.48***	0.48***	0.49***
	(0.02)	(0.02)	(0.02)
Cash 350	0.50***	0.50***	0.50***
	(0.02)	(0.02)	(0.02)
Cash 400	0.51***	0.51***	0.52***
	(0.03)	(0.02)	(0.02)
Cash 500	0.52***	0.52***	0.52***
	(0.02)	(0.02)	(0.02)
Female head		0.09**	0.08**
		(0.04)	(0.04)
Marginal respondent		0.08**	0.08**
		(0.04)	(0.04)
Round 2	0.05**	0.05*	0.05*
	(0.03)	(0.03)	(0.03)
Round 3	0.13***	0.13***	0.13***
	(0.03)	(0.03)	(0.03)
Constant	0.19***	0.17***	0.07
	(0.04)	(0.04)	(0.13)
Slum effects	Yes	Yes	Yes
HH/respondent controls	No	No	Yes
N	6507	6507	6480
R^2	0.20	0.21	0.22

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses are clustered at the household level. Respondent/household-level controls include: age of the respondent, binary variable =1 if the respondent is Hindu, binary variable =1 if the respondent belongs to a scheduled caste or scheduled tribe, proportion of literate members in the household, dependency ratio, the count of household assets, baseline monthly per capita expenditure of the household, binary variable =1 if the household owns their house, binary variable =1 if the household has a ration card, and binary variable =1 if a female redeemed the voucher.

Figure B3: Difference in the marginal effects of the cash amount on the probability of choosing cash (Female-headed minus Male-headed) (corresponds to Figure 6)



Note: Estimated from sample including male respondents.

C Additional Results

Table C1: Comparison between male-headed households and female-headed households restricted to female heads who are widowed/ separated/ with an absentee husband (corresponding to column 2 of Table 4)

Dependent Variable: DWL **Sample with only female respondents**	(1)
Female head	38.58*
	(23.45)
Marginal respondent	48.51**
	(22.58)
Round 2	29.12*
	(16.03)
Round 3	64.32***
	(15.32)
Constant	-128.05*
	(65.70)
Slum effects	Yes
Household effects	No
Respondent/household-level controls	Yes
N	589

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses are clustered at the household level. Respondent/household-level controls include: age of the respondent, binary variable =1 if the respondent is Hindu, binary variable =1 if the respondent belongs to a scheduled caste or scheduled tribe, proportion of literate members in the household, dependency ratio, the count of household assets, baseline monthly per capita expenditure of the household, binary variable =1 if the household owns their house, binary variable =1 if the household has a ration card, and binary variable =1 if a female redeemed the voucher.

D Conceptual Framework

Consider a household that makes a choice between receiving a cash benefit or the option to buy up to $\bar{x}_s$ units of rice at a subsidized price of p_s per unit. The market price is $p_m > p_s$.[18] We assume that market rice and subsidized rice are perfect substitutes,[19] and the household's total rice consumption x is non-zero, of which $x_s \leq \bar{x}_s$ is the amount of subsidized rice consumption. The household has a fixed budget of Y in a given time period, say, a month. We assume that in each household, the woman and the man of the household[20] have control over a certain part of the budget, but the woman is responsible for food spending. The woman controls a fraction $\alpha(b)Y$ of the budget, hence the man controls $(1 - \alpha(b))Y$. The parameter α is an increasing function of the woman's intra-household bargaining power, b. The expenditure for food is taken from the woman's budget.

For the choice between cash or in-kind transfer in this setup, there are two possibilities depending on whether the household is infra-marginal or marginal. For an infra-marginal household, $x > x_s$ and $x_s = \bar{x}_s$. For such a household, their total expenditure for rice is $E^R_{(1)} = p_m(x - \bar{x}_s) + p_s\bar{x}_s$, where $\bar{x}_s$ and $x - \bar{x}_s$ respectively denote the quantity of subsidized and market rice consumed by the household. For a marginal household, $x = x_s \leq \bar{x}_s$. Their rice expenditure is, $E^R_{(2)} = p_s x_s \leq p_s\bar{x}_s$.

If the woman is asked to decide whether to accept a cash transfer T or the option to buy $\bar{x}_s$ units of subsidized rice at p_s, she seeks to maximize $\alpha(b)Y - E^R$, i.e., her fraction of the budget left after food expenditures have been met. If she accepts the cash transfer, total household income increases by T but she must buy all rice at market prices. Assuming that bargaining power is constant, the cash transfer increases her budget by $\alpha(b)T$. In-kind benefit on the other hand does not expand her budget $\alpha(b)Y$, but lowers the amount she must spend on rice from her budget. If the household is infra-marginal, the amount she saves is $(p_m - p_s)\bar{x}_s$, the price difference between the market rice and subsidized rice, multiplied by the quantity of subsidized rice she can buy. If the household is marginal, she saves $(p_m - p_s)x_s$.

When deciding between in-kind benefits or cash transfers, the woman compares her savings from the subsidized rice against the budget expansion through a cash transfer. A woman from an infra-marginal household would be indifferent between subsidized rice and a cash transfer if $(p_m - p_s)\bar{x}_s = \alpha(b)T$. Hence, only if the value of the cash transfer T is greater than $T^*_{(1)} = \frac{(p_m - p_s)\bar{x}_s}{\alpha(b)}$, would she prefer cash over subsidized rice, where $T^*_{(1)}$ represents her willingness to pay for subsidized rice. If the household is marginal, $T^*_{(2)} = \frac{(p_m - p_s)x_s}{\alpha(b)}$. Since $x_s \leq \bar{x}_s$, it follows that $T^*_{(2)} \leq T^*_{(1)}$. In other words, the willingness to pay for subsidized rice is lower for a woman from a marginal household than from an infra-marginal household.

More importantly, both $T^*_{(1)}$ and $T^*_{(2)}$ are inversely related to $\alpha(b)$. Thus, regardless of

[18] In our experiment, p_s is set to zero.

[19] This is a reasonable assumption in the context of our experimental design since the subsidized rice we offered to the respondents was of comparable quality and sourced from the same local shops where the respondents bought their market rice.

[20] We use the terms "woman/man of the household" to refer to the key female or male figure in the household with decision-making authority, though such authority need not be equal.

marginality or infra-marginality, in cases where the woman has lesser (greater) control of the budget, her willingness to pay for subsidized rice is higher (lower), in turn implying a lower (higher) deadweight loss. This is because the woman realizes the full value of the subsidy if she receives subsidized rice, but only a fraction $\alpha(b)$ of it if she accepts the cash transfer.[21] This framework thus predicts that in male-headed households where $\alpha(b)$ is low, we could expect a greater likelihood of choosing rice over cash and lower deadweight loss (or possibly even deadweight gain), while in female-headed households with a high $\alpha(b)$, we could expect larger deadweight losses. These should however not be interpreted as welfare-enhancing gains or welfare-reducing losses for the entire household, since gains or losses accrue only to the decision-maker. A normative welfare evaluation for the *household* would need to aggregate gains and losses across all household members.

The conceptual framework assumes no intra-household transfers following the choice of rice or cash. However, insofar as there is some renegotiation of control over the budget over time, the difference between the choices of women from male- and female-headed households could be smaller. This is discussed further in Section 7.2.

[21] On the other hand, for the man of the household, choosing cash would expand his budget by $(1-\alpha)T$, while the subsidised rice option would leave his budget unchanged and only benefit the woman by lowering what she needs to spend for the total quantity of rice R consumed by the household.

E Experimental Instructions

STATIC TEXT

The following questions and the experiment has to be conducted on the same individual who was selected for the interviewin the last round. If that person is absent, identify any other adult.

Name of the respondent	TEXT selected ....
What is the age of %selected%?	NUMERIC: DECIMAL age _
What is the gender of %selected%?	SINGLE-SELECT gender 01 ○ Male 02 ○ Female
Is %selected% the head of the household?	SINGLE-SELECT head 01 ○ Yes 00 ○ No

[C] EXPERIMENT

STATIC TEXT

We will nowbe asking you to make choices between getting 5 kilos of rice or getting different amounts of cash. We will be asking you to make a choice nine times. Each time, we will be asking you to tell us whether you would prefer to get a particular amount of cash or 5 kilos of rice.

Please choose carefully because these are not just hypothetical choices. Later, one of these choices will become real when you drawa number from the lottery bag. The number you drawfrom the bag will tell us which of the nine choices is selected, and that will determine what you will get. For example, if you picked the number 300, then we will look at your preference between 300 rupees and 5 kilos of rice, and if you had chosen 5 kilos of rice, you will get 5 kilos of rice, not 300 rupees. Or instead, if you had chosen 300 rupees, you will get 300 rupees, not 5 kilos of rice.

So, your choices will matter to what you can get. So, please choose thoughtfully. So, let's nowbegin with asking you about your choices. Please note that at the current market price of about Rs. 32 per kilo of rice, the value of 5 kilos of rice is about 160 rupees.

[C] EXPERIMENT
CHOICE 500

STATIC TEXT

The image shows a sack of 5 kilograms of rice and 500 rupees in cash. Please look at the two images carefully and tell us which one do you choose.

Select one of the two options listed below	SINGLE-SELECT ch500 01 ○ I want 5 kilogram rice 02 ○ I want 500 rupees

[C] EXPERIMENT

CHOICE 400

STATIC TEXT

The image shows a sack of 5 kilograms of rice and 400 rupees in cash. Please look at the two images carefully and tell us which one do you choose.

Select one of the two options listed below	SINGLE-SELECT ch400 01 ○ I want 5 kilogram rice 02 ○ I want 400 rupees

[C] EXPERIMENT
CHOICE 350

STATIC TEXT

The image shows a sack of 5 kilograms of rice and 350 rupess in cash. Please look at the two images carefully and tell us which one do you choose.

Select one of the two options listed below	SINGLE-SELECT ch350 01 ○ I want 5 kilogram rice 02 ○ I want 350 rupees

[C] EXPERIMENT
CHOICE 300

STATIC TEXT

The image shows a sack of 5 kilograms of rice and 300 rupees in cash. Please look at the two images carefully and tell us which one do you choose.

Select one of the two options listed below	SINGLE-SELECT ch300 01 ○ I want 5 kilogram rice 02 ○ I want 300 rupees

[C] EXPERIMENT

CHOICE 250

STATIC TEXT

The image shows a sack of 5 kilograms of rice and 250 rupees in cash. Please look at the two images carefully and tell us which one do you choose.

Select one of the two options listed below	SINGLE-SELECT ch250 01 ○ I want 5 kilogram rice 02 ○ I want 250 rupees

[C] EXPERIMENT

CHOICE 200

STATIC TEXT

The image shows a sack of 5 kilograms of rice and 200 rupees in cash. Please look at the two images carefully and tell us which one do you choose.

Select one of the two options listed below	SINGLE-SELECT ch200 01 ○ I want 5 kilogram rice 02 ○ I want 200 rupees

[C] EXPERIMENT

CHOICE 150

STATIC TEXT

The image shows a sack of 5 kilograms of rice and 150 rupees in cash. Please look at the two images carefully and tell us which one do you choose.

Select one of the two options listed below	SINGLE-SELECT ch150 01 ○ I want 5 kilogram rice 02 ○ I want 150 rupees

[C] EXPERIMENT

CHOICE 100

STATIC TEXT

The image shows a sack of 5 kilograms of rice and 100 rupees in cash. Please look at the two images carefully and tell us which one do you choose.

५ किलो तांदूळ

१०० रुपये

Select one of the two options listed below	SINGLE-SELECT ch100 01 ○ I want 5 kilogram rice 02 ○ I want 100 rupees

[C] EXPERIMENT

CHOICE 50

STATIC TEXT

The image shows a sack of 5 kilograms of rice and 50 rupees in cash. Please look at the two images carefully and tell us which one do you choose.

५ किलो तांदूळ

५० रुपये

Select one of the two options listed below	SINGLE-SELECT ch50 01 ○ I want 5 kilogram rice 02 ○ I want 50 rupees

[D] EXPERIMENT RESULT

STATIC TEXT

Now we will draw a chit

What number came up in the lottery?	NUMERIC: INTEGER lottery _
The lottery resulted in	SINGLE-SELECT lotteryresult 01 ○ Cash 02 ○ Rice

What is the primary reason for choosing cash? E `lotteryresult==1`	SINGLE-SELECT `choicecsh` 01 ○ I chose cash because the cash amount is more than the value of 5 kilos of rice 02 ○ I chose cash because already have rice at home 03 ○ I chose cash because I can use it to buy other things or rice itself 04 ○ I chose cash because I can use it to buy a different quality of rice 05 ○ Others
What is the primary reason for choosing rice? E `lotteryresult==2`	SINGLE-SELECT `choicerice` 01 ○ I chose rice because the cash amount is less than the value of 5 kilos of rice 02 ○ I chose rice because cash will get spent on less useful things than rice 03 ○ I chose rice because we are running short of rice 04 ○ I chose rice because it is hard to control how cash will get spent 05 ○ Others
Who will go to the shop to collect cash/rice?	SINGLE-SELECT `shpcollect2` 01 ○ Me 02 ○ My husband 03 ○ My wife 04 ○ Other male family member 05 ○ Other female family member
In the last experiment, what did you win?	SINGLE-SELECT `lastoutcome` 01 ○ Cash 02 ○ Rice
In the last round, who went to the shop to collect cash/rice?	SINGLE-SELECT `shpcollect1` 01 ○ Me 02 ○ My husband 03 ○ My wife 04 ○ Other male family member 05 ○ Other female family member